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1. A shopkeeper sells 5 different products. Write a C program using a 2D array to store the quantity and unit price of each product. Calculate the total price of each product using quantity × price, display the quantity, unit price, and total price of each product, and calculate and display the grand total of all products using nested loops.

folderofc.c > C arraayyss.c

```c
#include <stdio.h>
int main() {
    int products[5][2];
    int i, j;
    int total;
    int grandTotal = 0;
    for (i = 0; i < 5; i++){
        printf("Enter quantity for product %d: ", i + 1);
        scanf("%d", &products[i][0]);

        printf("Enter unit price for product %d: ", i + 1);
        scanf("%d", &products[i][1]);
    }
    printf("\nProduct\tQuantity\tUnit Price\tTotal Price\n");
    for (i = 0; i < 5; i++){
        total = 0;
        for (j = 0; j < 1; j++) {
            total = products[i][0] * products[i][1];
        }
        grandTotal = grandTotal + total;
        printf("%d\t%d\t\t%d\t\t%d\n",
                i + 1,
                products[i][0],
                products[i][1],
                total);
    }
    printf("\nGrand Total = %d\n", grandTotal);
}
```

```
Enter quantity for product 1: 12
Enter unit price for product 1: 12
Enter quantity for product 2: 12
Enter unit price for product 2: 12
Enter quantity for product 3: 12
Enter unit price for product 3: 12
Enter quantity for product 4: 12
Enter unit price for product 4: 12
Enter quantity for product 5: 12
Enter unit price for product 5: 12

Product Quantity        Unit Price      Total Price
1       12              12              144
2       12              12              144
3       12              12              144
4       12              12              144
5       12              12              144
```

2. Write a C program using nested loops to print the following pattern: 1 1 4 1 4 27 1 4 27 16 1 4 27 16 125

```c
#include <stdio.h>

int main(){
    for(int i = 1; i <= 5; i++){
        for(int j = 1; j <= i; j++){
            if(j % 2 == 0){
                printf("%d ", j*j);
            }
            else{
                printf("%d ", j*j*j);
            }
        }
        printf("\n");
    }
}
```

```
1
1 4
1 4 27
1 4 27 16
1 4 27 16 125
PS C:\Users\I C\VS\folderofc.c>
```

3. Write a C program using nested loops to print the following pattern: 1 2 2 3 3 3 4 4 4 4 5 5 5 5 5

```c
#include <stdio.h>

int main(){
    for(int i = 1; i <= 5; i++){
        for(int j = 1; j <= i; j++){
            printf("%d",i);
        }
        printf("\n");
    }
}
```

```
1
22
333
4444
55555
PS C:\Users\I C\VS\folderofc.c>
```

4. Write a C program using nested loops to print the following pattern: 1 2 3 4 5 1 2 3 4 1 2 3 1 2 1

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```c
#include <stdio.h>

int main(){
    for(int i = 5; i >= 1; i--){
        for(int j = 1; j <= i; j++){
            printf("%d",j);
        }
        printf("\n");
    }
}
```

```
12345
1234
123
12
1
PS C:\Users\I C\VS\folderofc.c>
```

5. Write a C program that takes input for a 3×3 matrix using a 2D array. Apply appropriate conditions using nested loops to check and display whether the matrix satisfies the following types: 1. Square Matrix 2. Null Matrix 3. Diagonal Matrix 4. Upper Triangular Matrix

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```c
#include <stdio.h>
int main() {
    int a[3][3];
    int i, j;
    int null = 1;
    int diagonal = 1;
    int upper = 1;
    printf("Enter 9 values:\n");
    for(i = 0; i < 3; i++) {
        for(j = 0; j < 3; j++) {
            scanf("%d", &a[i][j]);
        }
    }
    for(i = 0; i < 3; i++) {
        for(j = 0; j < 3; j++) {
            if(a[i][j] != 0) {
                null = 0;
            }
        }
    }
    for(i = 0; i < 3; i++) {
        for(j = 0; j < 3; j++) {
            if(i != j && a[i][j] != 0) {
                diagonal = 0;
            }
        }
    }
    for(i = 0; i < 3; i++) {
        for(j = 0; j < 3; j++) {
            if(i > j && a[i][j] != 0) {
                upper = 0;
            }
        }
    }
    printf("\nMatrix Types:\n");

    printf("1. Square Matrix: Yes\n");

    if(null == 1)
        printf("2. Null Matrix: Yes\n");
    else
        printf("2. Null Matrix: No\n");

    if(diagonal == 1)
        printf("3. Diagonal Matrix: Yes\n");
    else
        printf("3. Diagonal Matrix: No\n");

    if(upper == 1)
        printf("4. Upper Triangular Matrix: Yes\n");
    else
        printf("4. Upper Triangular Matrix: No\n");

    return 0;
}
```

```
Enter 9 values:
2
4
6
8
10
12
14
16
18

Matrix Types:
1. Square Matrix: Yes
2. Null Matrix: No
3. Diagonal Matrix: No
4. Upper Triangular Matrix: No
```